

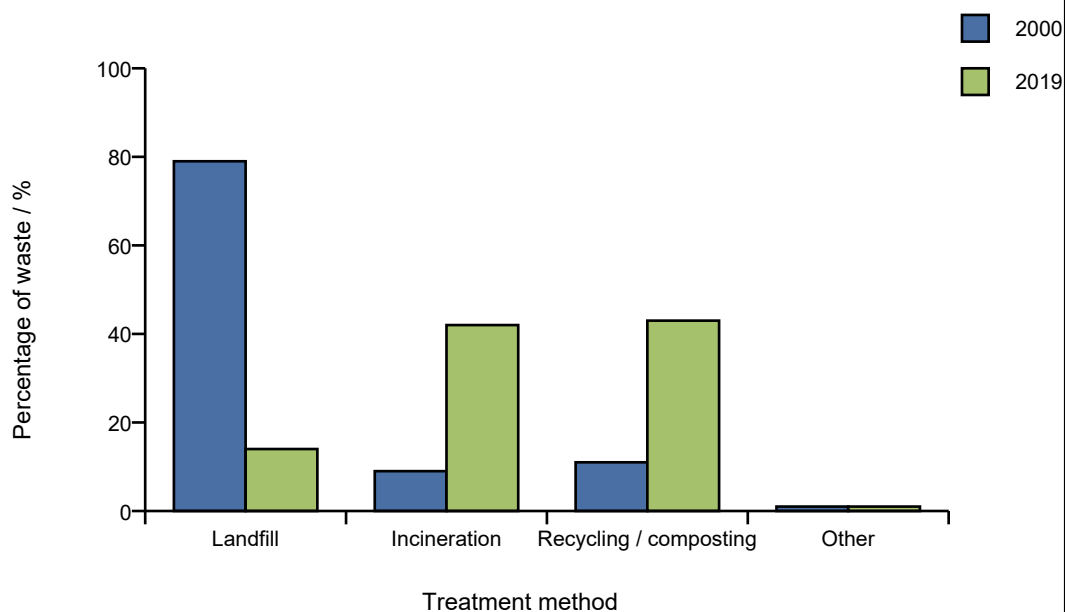
Questions by topic: Solid wastes - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1

Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 1

1

Describe the changes shown in Figure 1.

[2 marks]

.....

.....

.....

.....

0 1

2

Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

1

2

3

0 1

3

Describe **four** features of a well-managed landfill site.

[4 marks]

1

2

3

4

0 1 . 4

Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

.....

.....

10

0 2 . 1

Give **three** advantages of incinerating domestic waste.

[3 marks]

1

.....

2

.....

3

.....

0 2 . 2

Explain how each of the following features of a modern incinerator reduces pollution.

[4 marks]

High combustion temperature with rapid cooling of gases

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Water spray scrubber

.....

Removal of wet / compostable waste

.....

Removal of recyclable waste

.....

0 2 . 3 Explain why the ash from waste incinerators needs careful disposal. **[2 marks]**

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.....

.....

0 2 . 4 State **one** reason why incineration is expensive. **[1 mark]**

.....

.....

10

0 3 Table 1 shows three specialist solid wastes.

0 3 . 1 Complete Table 1 to show the main hazard of each waste and the disposal method used. **[6 marks]**

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

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0	3	.	2
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Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

0	3	.	3
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Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

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0	3	.	4
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Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

10

A coal-mine spoil heap on a hillside has been abandoned. The spoil contains shale with some iron pyrites and residual coal.

Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

10

0 5 . 1

Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

0 5 . 2

Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

1

2

3

0 5 . 3

Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage

Disadvantage

05 . 4

Explain why mixed municipal waste is difficult to recycle.

[1 mark]

.....

.....

10

06 . 1

Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients ☐ to the soil
- B It produces more electricity ☐
- C It removes heavy metals ☐
- D It destroys asbestos fibres ☐

06 . 2

Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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.....

06 . 3

Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

1

.....

2

.....

5

END OF QUESTIONS